

G. Lejeune Dirichlet's Werke. II.

XX.

Investigations on a problem of hydrodynamics

Translated from the German

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Investigations on a problem of hydrodynamics.

[Extract from a memoir presented to the Royal Society on 31 July 1857.]

Although the fundamental equations of hydrodynamics, in the form in which one finds them in textbooks, have been known since Euler, and in another form, due to Lagrange, since the appearance of the first edition of the *Mécanique analytique*, nevertheless in all cases investigated up to now in which the fluid changes its shape in the course of the motion, one has not derived the complete solution of the problem from the fundamental equations, but has confined oneself to an approximate determination of the motion. Prompted by the circumstance just mentioned to attempt to treat a problem of the indicated kind in complete rigor, the author of the memoir, given the difficulty of the subject, whose resolution requires the integration of a system of nonlinear partial differential equations, was prepared for such an attempt to succeed only under the simplest hypotheses on the conditions determining the motion of the fluid. He was therefore not a little surprised when, after several fruitless efforts, a case presented itself which is not to be counted among the simpler ones, insofar as the mutual attraction of the elements of the fluid is taken into account, and in which, notwithstanding this circumstance, the motion can be recognized not only in its general character, but also, under certain restrictions, determined in its details.

The conditions of this case of motion, which is treated in the memoir, and the general result relating to it, are expressed in the following proposition:

If a homogeneous incompressible fluid which at its surface undergoes a constant pressure, or one varying only with time, initially has the shape of an ellipsoid; and if, moreover, the initial motion is decomposable into two simpler motions, a rotation of the fluid like that of a rigid body about an axis passing through the center of mass, and a second motion changing the relative position of the elements, in which the velocities of those elements, resolved perpendicular to three fixed planes meeting at right angles at the center, are proportional to the distances from the planes, then the fluid, in the motion arising as a consequence of such an initial state, will at every later time also have the shape of an ellipsoid concentric with the original one, but whose axes in general change with time both in direction and in magnitude. Of the instantaneous motion occurring at any given time, the same holds as was assumed for the original motion; that is, it is decomposable into two simpler motions of the kind just defined, but so that both the axis of rotation and the three mutually perpendicular planes to which these component motions refer also in general assume a different position at every instant.

For complete knowledge of the motion whose general character has just been indicated, it is necessary to determine 9 functions of time, which are defined by just as many equations,

one finite equation and 8 differential equations of the second order. For the latter one can in general set up 7 first integrals; these, however, are not sufficient for the complete solution of the problem, that is, for its reduction to quadratures, unless the initial shape and the state of motion occurring in the first instant, which contains 8 arbitrary elements, are subjected to further restrictions.

Among the cases reducible to quadratures, the simplest, and the only one that can be mentioned here, is the case in which the mass originally has the form of a spheroid and no initial velocities are present. The motion then consists of isochronous oscillations in which the fluid, passing through the spherical shape, alternately assumes the form of a prolate and of an oblate ellipsoid.